

SPENCER NG

Redwood City, CA · sng@spencerng.me · github.com/spencerng
Creative Technologist · Live Experience Designer

EDUCATION

The University of Chicago

M.S. & B.S. in Computer Science, *summa cum laude*
B.A. in Theater & Performance Studies with honors

Chicago, IL
Graduated June 2023
GPA: 3.96/4.00

WORK EXPERIENCE

Roblox

Software Engineer

San Mateo, CA

Aug. 2023 - Present

- Ensured stable release of rewarded video ads by leading A/B testing, working with executive and product stakeholders, and creating SQL Airflow pipelines to detect regressions
- Scaled video .NET microservices for video transcoding, safety scanning, and cost-to-serve metrics to enable longer UGC video uploads and handle 10M requests/day with >99.99% service reliability
- Optimized video playback quality, stability, and memory consumption on the game engine
- Processed voice quality metrics at scale by optimizing and deploying Triton ML model servers

Software Engineering Intern

Jun. 2022 - Sep. 2022

- Improved audio store search by building a production gRPC .NET service to classify uploaded sounds
- Detected copyrighted audio clips and made SFX recommendations to users by researching, developing, and deploying audio deep learning models and vector similarity search on AWS Lambda

University of Chicago Computer Science Department

Teaching Assistant

Chicago, IL

Sep. 2020 - Jun. 2023

- Increased grading efficiency and accuracy by building scalable Python infrastructure for autograding student Haskell, C, and Python code via Gradescope and GitHub Classroom
- Designed assignments on VR activity detection in Unity and electronic chip counters

Verizon

Product Management Intern

Basking Ridge, NJ

Jun. 2021 - Aug. 2021

- Defined product vision, UX flow, and feature requirements for mobile experiences integrating AR and tracking technologies in partnership with NBA and NFL teams
- Improved user experience and decreased AR load times in the NFL 5G Multi-View app by presenting recommendations to product vendors and owners

University of Chicago IT Services

Network Architecture Developer

Chicago, IL

Oct. 2019 - Jun. 2021

- Automated campus network maintenance by creating Python interfaces to determine real-time device status and backup/sync device configurations between Netbox, Box, and local storage
- Implemented and documented Docker/Gitlab CI workflow for Python development and deployment

University of Chicago Laboratory Schools

Metcalf Intern

Chicago, IL

Apr. 2020 - Jun. 2020

- Created dashboards with Tableau to analyze the effectiveness of learning management and video conference platforms

SKILLS

Languages: Python, C#, SQL, JavaScript, HTML/CSS, Java, C, C++, Unix Bash, Processing
Tools: Git, AWS, PyTorch, Docker, Superset, Grafana, Prometheus, Unity, Android, ROS, Twilio
Other: OBS, QLab, 3D printing/modeling, Adobe Creative Suite, puzzle design, digital electronics

PUBLICATIONS

Spencer Ng, Ting-Han Lin, You Li, and Sarah Sebo (2024). Role-Playing with Robot Characters: Increasing User Engagement through Narrative and Gameplay Agency. In *Proceedings of the 2024 ACM/IEEE International Conference on Human-Robot Interaction (HRI 2024)*.

Ting-Han Lin*, **Spencer Ng***, and Sarah Sebo (2022). Benefits of an Interactive Robot Character in Immersive Puzzle Games. In *Proceedings of the 31st IEEE International Conference on Robot & Human Interactive Communication (RO-MAN 2022)*. | *equal contribution

RESEARCH EXPERIENCE

Human-Robot Interaction Lab, University of Chicago

Chicago, IL

Research Assistant, advised by Prof. Sarah Sebo

Oct. 2020 - Mar. 2024

- Published research on how robot characters in immersive entertainment settings are more fun to play with than humans and can increase user engagement by creating agency
- Increased programming efficiency by writing open-source Robotics Operating System APIs

Argonne National Laboratory

Lemont, IL

Research Intern, advised by Dr. Nicola Ferrier

Jun. 2020 - Sep. 2020

- Determined the most bandwidth-optimized samples for transfer learning by designing a Python pipeline to run inference on simulated edge sensors with custom-trained PyTorch models

Neural Engineering Speech and Hearing Lab, NJ Institute of Technology

Newark, NJ

Research Intern, advised by Prof. Antje Ihlefeld

Jun. 2019 - Sep. 2019

- Diagnosed hearing disabilities by developing a sound matching Unity game with C#, incorporating custom real-time algorithms to simulate pitch shifting and sound localization

LEADERSHIP EXPERIENCE

University Theater

Chicago, IL

Committee Chair & Stage/Production Manager

Oct. 2019 - Jun. 2023

- Produced 20+ annual shows by managing a \$50k+ budget and curating diverse/student-written work
- Foster artistic opportunities, resolve conflicts, and create inclusive rehearsal spaces for 100+ members
- Maintained University Theater online presence and Jekyll-based website, with 5,000+ monthly views

Uncommon Hacks

Chicago, IL

Co-Director

Jan. 2021 - Jun. 2022

- Hosted an in-person hackathon and workshop series for 120+ students by leading a team of 15+ organizers, managing food and prize logistics, and raising \$8k+ in sponsorship funds
- Created and led a workshop on real-world data visualization using JavaScript and Chart.js
- Managed judges/mentors and created social events for an online hackathon with 300+ attendees

UChicago Science Olympiad

Chicago, IL

Event Supervisor

Oct. 2019 - Jun. 2021

- Created and administered hands-on tests for high school science competitions with 150+ annual participants, focusing on electronic circuit design/analysis and technical communication skills

Magic

Chicago, IL & New Jersey

Performer & Instructor

Sep. 2017 - Jun. 2023

- Entertain crowds of up to 200 at community events, comedy clubs, and birthday parties
- Wrote a curriculum and taught a 20-week course on sleight-of-hand magic to children ages 10+

SELECTED PROJECTS

Learning Institute for the Very Enlightened: Immersive narrative escape room with custom-made monitoring/cueing system, Arduino automation, smart light control, projected media, and physical puzzles

Conversational AI Characters: Interactive art installation featuring iconic robot characters in endless conversation, built with GPT-3 prompt engineering, 3D-printed parts, and custom text-to-speech models

Cloud Carbon Estimation: Browser extension integrated in AWS and GCP consoles to estimate carbon emissions caused by launching virtual machines via ClimaTiq API data

Embodied Controllers: Installation to play emulated retro console games using physical gestures (e.g. jumping), built using Azure Kinect, OpenCV, OBS, and PyGame

AutoTOS: Website to summarize privacy policy terms, built by training a natural language processing model (90% accuracy) and writing a Google Cloud backend (Best Use of Google Cloud, PennApps Fall '20)

AWARDS & HONORS

Theater & Performance Studies Award: BA project with the highest level of artistic excellence 2023

Phi Beta Kappa: academic honor society 2022

Enrico Fermi Scholar: top 5% GPA in the Physical Sciences Collegiate Division 2022

Harper Award for Exceptional Performance in a Course: awarded for Mobile Computing 2021

TEACHING EXPERIENCE

University of Chicago

Teaching Assistant

Mobile Computing	Winter 2022, Winter 2023, Spring 2023
Transmedia Puzzle Design & Performance	Fall 2022
Introduction to Computer Science I	Fall 2022
Computer Science for Data Scientists	Spring 2022
Honors Introduction to Computer Science I	Fall 2021
Engineering Interactive Electronics onto Printed Circuit Boards	Spring 2021

Course Grader

Computer Science with Applications III	Spring 2021
Honors Introduction to Computer Science II	Winter 2021
Honors Introduction to Computer Science I	Fall 2020

THEATER PRODUCTION EXPERIENCE

Regional Theater

<i>How I Learned What I Learned</i> , Asst. Projections Designer, TheatreWorks Silicon Valley	Winter 2024
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University of Chicago

<i>Learning Institute for the Very Enlightened</i> , Writer/Director/Designer	Spring 2023
<i>At Your Own Risk</i> , Asst. Projections Designer, Theater & Performance Studies	Spring 2023
<i>Macbeth in Space</i> , Projections Designer, University Theater	Winter 2023
<i>The Intruder</i> , Projections Designer, University Theater	Fall 2022
<i>Theater[24]</i> , Actor, University Theater	Fall 2022
<i>King Lear</i> , Production Manager, University Theater	Spring 2022
<i>Amazons and their Men</i> , Asst. Scenic Designer, Theater & Performance Studies	Winter 2022
<i>Love's Labour's Lost</i> , Stage Manager, University Theater	Fall 2021
<i>My H8 Letter to the Gr8 American Theatre</i> , Stage Manager, Theater & Performance Studies	Winter 2021
<i>Waiting for Godot</i> , Asst. Stage Manager, University Theater	Winter 2020
<i>The Winter's Tale</i> , Asst. Stage Manager, University Theater	Fall 2019